

Circuit Simulation Lab:27

Design of memories

The purpose of this experiment is to design a single Port 8 bit RAM Synchronous Read/Write.

CODE:

```
module single_port_ram
#(
parameter DATA_WIDTH = 8,
parameter ADDR_WIDTH = 4
)
(
input clk,
input c_en,
input wr_e,
input out_e,

input [ADDR_WIDTH-1:0] addr,
input [DATA_WIDTH-1:0] data,

output reg [DATA_WIDTH-1:0] q
);

reg [DATA_WIDTH-1:0] mem [0:(2**ADDR_WIDTH)-1];

always @(posedge clk)
begin

if(c_en)
begin

if(wr_e)
mem[addr] <= data;

else if(out_e)
q <= mem[addr];

end

end

end
```

```
Endmodule
```

```
TESTBENCH
```

```
module tb_single_port_ram;
```

```
reg clk;
```

```
reg c_en;
```

```
reg wr_e;
```

```
reg out_e;
```

```
reg [3:0] addr;
```

```
reg [7:0] data;
```

```
wire [7:0] q;
```

```
single_port_ram dut(
```

```
.clk(clk),
```

```
.c_en(c_en),
```

```
.wr_e(wr_e),
```

```
.out_e(out_e),
```

```
.addr(addr),
```

```
.data(data),
```

```
.q(q)
```

```
);
```

```
always #10 clk = ~clk;
```

```
initial begin
```

```
$dumpfile("ram.vcd");
```

```
$dumpvars(0,tb_single_port_ram);
```

```
clk = 0;
```

```
c_en = 1;
```

```
// -----
```

```
// WRITE OPERATIONS
```

```
// -----
```

```
wr_e = 1;
```

```
out_e = 0;
```

```
addr = 4'd0;
```

```
data = 8'hAA;  
#20;
```

```
addr = 4'd1;  
data = 8'h55;  
#20;
```

```
addr = 4'd2;  
data = 8'hF0;  
#20;
```

```
// -----  
// READ OPERATIONS  
// -----
```

```
wr_e = 0;  
out_e = 1;
```

```
addr = 4'd0;  
#20;
```

```
addr = 4'd1;  
#20;
```

```
addr = 4'd2;  
#20;
```

```
$finish;
```

```
end
```

```
initial begin  
$monitor(  
"time=%0t addr=%d data=%h q=%h wr_e=%b",  
$time,addr,data,q,wr_e  
);  
end
```

```
endmoduleA
```

```

vlsi_dev_box@vlsi-box:~/Documents/NIELIT_ASSIGNMENTS/EXP_NO_27$ iverilog -o out
single_port_ram.v tb_single_port_ram.v
vlsi_dev_box@vlsi-box:~/Documents/NIELIT_ASSIGNMENTS/EXP_NO_27$ vvp out
VCD info: dumpfile ram.vcd opened for output.
time=0 addr= 0 data=aa q=xx wr_e=1
time=20 addr= 1 data=55 q=xx wr_e=1
time=40 addr= 2 data=f0 q=xx wr_e=1
time=60 addr= 0 data=f0 q=xx wr_e=0
time=70 addr= 0 data=f0 q=aa wr_e=0
time=80 addr= 1 data=f0 q=aa wr_e=0
time=90 addr= 1 data=f0 q=55 wr_e=0
time=100 addr= 2 data=f0 q=55 wr_e=0
time=110 addr= 2 data=f0 q=f0 wr_e=0

```

